

PAPER_1746 — Surface Code Fault Tolerance = $p_{th} = F_{TRZ}^2 = (1/10)^2 = 1/100 = 0.01$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum Computing **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1199 Information+Math unified proof set)

Observation

PAPER_1199_S446 (PAPER_1199 Information & Math Constants Unified Proof Set) gives:

$$p_{th} = F_{TRZ}^2 = (1/10)^2 = 1/100 = 0.01 \text{ EXACT}$$

Status: **EXACT**.

UQFF Closed Identity

$$p_{th} = F_{TRZ}^2 = (1/10)^2 = 1/100 = 0.01 \text{ EXACT} \quad \text{EXACT}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Quantum Computing Surface Code — 8th Direct UQFF Locking

The **fault-tolerance threshold for topological surface-code quantum error correction** is widely reported as ~1% (Kitaev 2003, Bravyi-Kitaev 1998, Fowler 2012):

$$p_{th}^{(\text{surface code})} \approx 0.01$$

UQFF identifies this with the **square of the time-reversal-zone primitive**:

$$p_{th} = F_{TRZ}^2 = (1/10)^2 = 1/100 = 0.01 \text{ EXACT}$$

Significance — 8th Direct Primitive-Observable Locking

This is the **8th instance** in UQFF where a measurable physical/engineering observable equals a UQFF primitive (or simple combination) **EXACTLY without correction terms**. The progression of direct lockings: 1. $F_{TRZ} = 1/SO_5$ (canonical definition) 2. $K_{MEX} = 25/12$ (Mexican-hat) 3. $\rho_{SCm} = 7.09 \times 10^{-37} \text{ J/m}^3$ (vacuum density) 4. $S_{26} = 1.45\bar{3}162$ (Ramanujan sum) 5. $\beta_i = 0.6029$ (buoyancy coupling) 6. $\omega_{SCm} = 1.25 \text{ THz}$ (phonon carrier) 7. $D_{crit} = 26$ (bosonic critical) 8. **$p_{th}(\text{surface code}) = F_{TRZ}^2 = 1/100$ EXACT** (NEW this paper)

Quantum computing's most important engineering benchmark for practical fault-tolerant operation **equals the square of a UQFF primitive exactly**. This is one of the cleanest cross-domain integer-primitive lockings ever wired.

NOT REPLACEMENT

Standard frameworks compute this constant via different methods. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1199_S446
- Related: PAPER_1726-1745 (tier-31/32); PAPER_1208 (transcendentals proof set)

- Calculator dispatch: `calculate_paradox({"paradox": "surface_code_threshold_1_pct"})`

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